Case 3

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Status	Cases

1. How biochemical models of plasticity are built

Biochemically detailed models write each signalling step as a chemical reaction; the reactions become ordinary differential equations (ODEs) for every species' concentration, and simulating them shows how a stimulus pattern travels to a plasticity readout.

- **Building blocks**
  - Bimolecular reaction (mass action): A + B ⇌ C + D; rate ∝ product of reactant concentrations; forward rate kf, backward rate kb
  - Enzymatic reaction (Michaelis–Menten): E + S ⇌ ES → E + P; kcat = speed of the catalytic step; Km = (kb + kcat)/kf = how much substrate is needed for half-maximal speed
  - Cooperative binding (several ligands, each easing the next) → Hill function; e.g. cAMP binding to PKA
- **Deterministic vs stochastic**
  - Deterministic ODE: assumes one well-mixed compartment and enough molecules for mass action
  - Stochastic (Gillespie algorithm, τ-leap, chemical Langevin equation): needed when molecule numbers are small (a spine) → random fluctuations matter
  - Reaction–diffusion: needed when location matters (microdomains, anchored enzymes)
- **Phenomenological vs biochemically detailed**
  - Phenomenological rules (BCM: small Ca^{2+} → LTD, large Ca^{2+} → LTP; STDP rules) describe outcomes, not molecules → cannot say what a drug,

mutation or protein level does

    - Biochemically detailed models → can test which protein, pathway or neuromodulator a form of plasticity depends on

- **Choice of readout can decide the conclusion**
    - Cumulative activity = area under the curve of active enzyme → total substrate phosphorylated (a kinase active for 60 s does more than one active for 10 s)
    - Peak or single-time-point activity can hide or reverse differences between protocols (see §3)
    - Peak is appropriate when the product is protected from reversal (e.g. phosphoCaMKII anchored at the PSD)
- **Robustness (sensitivity) analysis**
    - Vary the least-constrained parameters (enzyme amounts, rates) and check whether the qualitative result survives
    - Sensitivity = relative change in output / relative change in a parameter → shows which parameters control the behaviour

. The signalling modules the models share

All the plasticity models are built from the same three kinase modules — PKA, CaMKII and PKC — driven by Ca^{2+} and neuromodulators and opposed by phosphatases and, for cAMP, phosphodiesterases (PDEs). Only what goes beyond earlier notes (Aplysia cAMP–PKA, AC as coincidence detector, CaMKII switch, calcineurin LTD, cerebellar PKC–AMPAR endocytosis) is spelled out here.

cAMP–PKA module

Turns coincident activity (Ca^{2+}) and neuromodulator (Gs-coupled receptors: dopamine D1, β-adrenergic) into a kinase signal.

1. NMDAR → Ca^{2+} influx → Ca^{2+} binds calmodulin (CaM); four Ca^{2+} = active $CaMCa_4$
2. $CaMCa_4$ + Gsα-GTP bind adenylyl cyclase AC1 synergistically = coincidence detector → ATP converted to cAMP (AC8: Ca^{2+}/CaM only)

3. cAMP binds the PKA holoenzyme (2 regulatory + 2 catalytic subunits); four cAMP bind cooperatively
4. Fully cAMP-bound regulatory subunits release the catalytic subunits = active PKA
5. Active PKA phosphorylates its targets (below)
6. PDEs hydrolyse cAMP → 5′AMP (the only enzymatic cAMP removal) → catalytic subunits reassociate → PKA off

- PDE1 is itself activated by Ca^{2+}/CaM → Ca^{2+} both makes and breaks cAMP; PDE4 = the main cAMP-specific PDE (§4)
- Active PKA is scarce: regulatory subunits keep binding catalytic subunits tightly even with four cAMP bound
    - Possible amplifiers: the cAMP-saturated holoenzyme may itself be active; AKAPs anchor PKA next to its substrates → few molecules suffice locally

PKA targets relevant here

- GluR1 S845 → faster AMPAR insertion → LTP (§5)
- Inhibitor-1 = the PKA gate on phosphatase:
    1. PKA phosphorylates inhibitor-1 (I-1)
    2. Phospho-I-1 binds and inhibits PP1
    3. Less PP1 → less dephosphorylation of CaMKII and other plasticity proteins → their phosphorylation persists
    4. Calcineurin (PP2B, Ca^{2+}/CaM) dephosphorylates I-1 → PP1 free again (the low-Ca^{2+} LTD route)

CaMKII: what the models add

- Calmodulin is shared and limiting
    - CaM is bound with high affinity by calcineurin and PDE1, with intermediate affinity by AC1 and CaMKII → each train finds less free CaM
    - Consequence in a closed compartment: later trains give smaller CaMKII increments → the CaMKII positive-feedback loop disappears

    - Restored by CaM replenishment: diffusion from the dendrite into the spine + neurogranin (releases CaM on Ca^{2+} = CaM reservoir)
    - CaM depletion also limits AC1 → limits cAMP (§3)
- CaM activation is slow and needs four Ca^{2+} → CaMKII needs large Ca^{2+} signals (§5)

PKC module (Gq route)

Lets glutamate (mGluR) and acetylcholine (M1), not only Ca^{2+}, drive AMPAR removal.

1. mGluR or muscarinic M1 receptor → activates Gq
2. Gq + Ca^{2+} activate PLC → PLC cleaves PIP_2 → DAG (+ IP_3)
3. Ca^{2+} + DAG activate PKC = transiently active PKC
4. Ca^{2+}-activated PLA2 → arachidonic acid → converts PKC into a persistently active form
5. PKC phosphorylates GluR2 S880 → GluR2 endocytosis (as in cerebellar LTD)
6. PP2A dephosphorylates S880 → GluR2 can return to the membrane

3. Temporal sensitivity: why spaced L-LTP needs PKA

L-LTP induced by four tetani needs PKA only when the trains are spaced (minutes apart), not massed (seconds apart). The model explains this as two kinases with opposite temporal sensitivity: spaced trains give more cumulative PKA but less CaMKII, so PKA must compensate.

- Same total stimulation in both protocols → only the temporal pattern differs

Why spaced trains activate more PKA

Spacing lets the cAMP machinery recover between trains, so each train counts fully.

1. Train 1: Ca^{2+}/CaM + Gs (dopamine released by the tetanus) activate free AC1 → cAMP pulse → PKA

2. Massed: train 2 arrives while AC1 and CaM are still bound/depleted → smaller AC1 increment → sublinear summation of cAMP
3. Massed PKA: higher peak, but later trains barely raise it → total PKA far below four times one train
4. Spaced: cAMP falls back to baseline within tens of seconds → AC1 and CaM recover → each train gives a full response
5. Spaced PKA: lower peak, but contributions add up linearly over time → higher cumulative PKA

- Peak vs cumulative: massed wins on peak, spaced on area → only the cumulative measure shows the temporal sensitivity
    - A model reading PKA at one time point saw none → readout choice decides the conclusion (§1)
- Cumulative PKA rises steeply with interval and saturates quickly (within tens of seconds)

CaMKII has the opposite sensitivity

- Massed trains summate before phosphoCaMKII decays → higher phosphoCaMKII
- Longer intervals → phosphoCaMKII declines exponentially
- Normalised PKA + CaMKII ≈ constant across intervals → PKA compensates for the CaMKII deficit
    - = several kinases, each tuned to a different temporal pattern, jointly cover all patterns
- Critical interval: CaMKII has fallen to its low level after ~3 time constants ≈ 1 min → prediction: PKA required for intervals above ~1 min

Downstream and dopamine

- Spaced → more phospho-I-1 → stronger PP1 suppression → protects phosphorylation of plasticity proteins
    - But blocking only I-1 phosphorylation barely reduced CaMKII → other PKA routes (GluR1 S845, anchored PKA) must contribute

- Blocking dopamine D1 receptors lowers PKA for both protocols
    - Spaced without dopamine ≈ massed with dopamine → spaced L-LTP needs dopamine-driven PKA
    - Massed L-LTP predicted insensitive to D1 block (enough CaMKII)

Evidence: mouse hippocampal slices (Schaffer–CA1), four tetani at 40 s or 80 s intervals with the PKA inhibitor KT5720 vs vehicle. 80 s L-LTP was blocked, 40 s L-LTP was unaffected → confirms the predicted critical interval between them.

4. PDE4 isoforms: feedback control of cAMP

PDE4 isoforms (long, short, supershort) all hydrolyse cAMP but differ in regulatory domains, so PKA and ERK — both activated downstream of cAMP — feed back on them in opposite ways. These feedbacks only matter when cAMP is sustained or oscillating, and then long isoforms dominate cAMP control.

- Isoforms arise from the four PDE4 genes via alternative promoters and splicing; the N-terminal regulatory regions set both regulation and subcellular location

Isoform-specific feedback rules

- Long: PKA phosphorylation → activated; ERK phosphorylation → inhibited; both → PKA wins (relieves ERK inhibition)
- Short: no PKA-regulated domain; ERK → activated
- Supershort: no PKA-regulated domain; ERK → inhibited

The ERK arm (cAMP → Epac → ERK)

Gives cAMP a second, slower feedback route onto PDE4.

1. cAMP binds Epac → releases Epac's autoinhibition (needs more cAMP than PKA)
2. Active Epac activates Rap1 (GDP → GTP)
3. Rap1 → B-Raf → MEK → dual phosphorylation of ERK (neurons)
4. Phospho-ERK dimerises → the dimer is the form that phosphorylates cytosolic PDE4

→ ERK rises more slowly than PKA

What each loop does

- PKA → long PDE4 activation = fast negative feedback: more cAMP → more PKA → faster cAMP removal → stabilises cAMP
- ERK → short PDE4 activation = slower negative feedback
- ERK → long/supershort inhibition = positive feedback: cAMP → ERK → less degradation → more cAMP

Model findings

- Single cAMP pulse: identical degradation for every isoform; only the total PDE4 amount matters
  - PKA/ERK rise too little and too late to change PDE4 activity
- Oscillatory or sustained cAMP input: continuous PKA/ERK activation → feedback engaged
  - Long only → cAMP stabilises (PKA activation outweighs ERK inhibition)
  - Short only → stabilises, but after a larger initial peak (ERK activation is slow)
  - Supershort only → ERK progressively shuts it off → runaway cAMP accumulation → other mechanisms needed (e.g. transcriptional upregulation, other PDEs)
- Inhibiting long isoforms raises cAMP, PKA, Epac and ERK most — also when all isoforms are equally abundant → not just an abundance effect
  - Short vs supershort importance depends on their relative amounts
- Pre-existing activity primes later signals
  - Pre-active PKA speeds degradation and overrides pre-active ERK
  - Pre-active ERK alone slows degradation (inhibits long and supershort) = permissive for later cAMP signals
- Raising cAMP ≠ lowering PDE4

    - PKA more sensitive to the amount of cAMP input; Epac and ERK more sensitive to the amount of PDE4
    - Changing PDE4 also shifts timing (time to peak); changing cAMP input mainly shifts amplitude
    - Responses are non-linear: larger cAMP inputs speed their own degradation (via PKA → long PDE4)
- Relevance: PDE4 inhibitors raise cAMP but cause side effects (nausea, emesis) → isoform-selective (long-form) targeting

Limits of the well-mixed model

- Pulses are degraded far faster than in live cells → real cAMP signalling is compartmentalised
    - Localised PDEs, AKAP complexes (PKA, Epac, PDE4 together), cAMP buffering by PKA regulatory-subunit condensates → nanometre-scale cAMP domains

5. Unified model of cortical post-synaptic plasticity

One spine model combines the CaMKII, PKA and PKC modules (§2) and converts their activity into AMPAR subunit traffic: PKA and CaMKII add and strengthen GluR1 (LTP), PKC removes GluR2 (LTD). Because the pathways need different amounts of Ca^{2+} and different neuromodulators, the model produces BCM-like frequency rules and neuromodulator-gated STDP.

Different Ca^{2+} thresholds for the three pathways

- Most Ca^{2+} entering the spine is bound at once by buffers and pumps → only a small free fraction
- PKC pathway proteins (PLC, PLA2, DAG lipase) bind Ca^{2+} fast and need one Ca^{2+} each → active at low Ca^{2+} (given mGluR/M1 ligand)
- CaM needs four Ca^{2+} and activates slowly → stays largely unbound at low Ca^{2+}
    - → PKA (via CaM–AC1, plus β-adrenergic ligand) needs relatively large Ca^{2+}

    - → CaMKII needs the largest Ca^{2+}
- Threshold order: PKC < PKA < CaMKII → basis for low Ca^{2+} → LTD, high Ca^{2+} → LTP

AMPAR readouts

- GluR1 S845 (PKA) → faster membrane insertion → more receptors
- GluR1 S831 (CaMKII; also PKC) → higher single-channel conductance of GluR1 homomers
- GluR2 S880 (PKC) → endocytosis

From subunit numbers to synaptic strength

Problem: too many subunit-state combinations to simulate tetramer assembly → estimate strength statistically.

- Assume random assembly: P(GluR1 homomer) ≈ (GluR1 fraction)^4, P(GluR2 homomer) ≈ (GluR2 fraction)^4, the rest heteromers
- Total conductance = number of each tetramer type × its single-channel conductance
    - GluR1 homomers: large (larger still with S831 phosphorylated); GluR2-containing: small
- Consequence: strength depends steeply on the GluR1 fraction
    - GluR2 endocytosis while GluR1 is scarce → fewer receptors → LTD
    - GluR2 endocytosis while GluR1 is being added → more GluR1 homomers → contributes to LTP (blocking PKC reduced HFS-LTP)
- Conclusions hold with an alternative assembly rule (dimers of like dimers)

Frequency-dependent plasticity

- HFS: large Ca^{2+} → CaMKII + PKA (with β-adrenergic input) → GluR1 insertion and S831; PKC → GluR2 removal → net LTP
- LFS: small Ca^{2+} → CaM not activated → no CaMKII/PKA; PKC still active → GluR2 removal → LTD

- Gating: LTP requires β-adrenergic ligand; LTD requires mGluR or M1 activation
- Both subunits are needed: no GluR1 → no LTP; no GluR2 → no LTD

Neuromodulator-gated STDP

Neuromodulators decide which side of the STDP curve a synapse can express.

1. Multicompartmental L2/3 pyramidal neuron model: presynaptic input to a spine paired with postsynaptic spikes at different intervals
2. Back-propagating spikes depolarise the spine → relieve the NMDAR Mg^{2+} block → Ca^{2+} influx depends on pre–post timing
3. That Ca^{2+} time course drives the spine biochemistry model
4. Outcome by modulator:
    - None → little change
    - β-adrenergic only → LTP window for pre-before-post
    - Acetylcholine (M1) only → LTD at most intervals
    - Both → full bidirectional STDP (LTP pre→post, LTD otherwise)
- Total Ca^{2+} over the pairing period predicts the outcome much better than peak Ca^{2+}
- SK channels shape the LTD side:
    1. More postsynaptic spikes per burst → more Ca^{2+}-activated SK current, decaying slowly
    2. SK hyperpolarises the neuron → less NMDAR Ca^{2+} from a presynaptic input arriving afterwards
    3. → stronger LTD for post-before-pre; fewer spikes or SK block → shorter LTD window, smaller LTD

Evidence: visual cortex L2/3 slices, pre–post pairing at different intervals with neuromodulator agonists: β-adrenergic activation alone gave LTP, muscarinic alone gave LTD, both together gave the classic bidirectional STDP curve → neuromodulators set STDP polarity.

Same rule, many protein combinations; different rules, different proteins

- Randomly varied protein concentrations → many different combinations give the same plasticity curve (degeneracy)
- Decisive parameters
  - GluR1 fraction: low → LTD only; intermediate → BCM-like; high → strong LTP
  - Ca^{2+} extrusion (NCX): fast extrusion limits LTP
  - PKA-pathway amount: strong LTP requires it; PKC-pathway amount matters for low-Ca^{2+} outcomes
- Plasticity differences between cortical areas and cell types can come from protein expression alone (reaction rates fixed)
  - Fitting only concentrations reproduced LTP/LTD data from many cortical areas; fitted parameters matched each study's proposed pathway (CaMKII-dependent vs PKA-dependent inputs to entorhinal cortex)
- Predictions
  - Nearly all cortical post-synaptic LTP is PKA-dependent
  - PKC block turns LTD into LTP → experiments should manipulate several pathways
- Robust: small parameter changes never abolished LTP or LTD; most influential were Ca^{2+} extrusion, PP1, AC1, PKA and GluR1 insertion

6. Reproducibility, replicability and comparability

A model's result is only trustworthy and reusable if others can obtain it again — by rerunning the authors' code (replication) or by rebuilding the model from the paper (reproduction) — and if models can be compared on equal terms.

Definitions (as used here; terminology is contested)

- Replicability: rerun the authors' public code → get the original results

- Reproducibility: reimplement the model from the publication (often in another tool) → get the original results
- Comparability: same model in different tools → same results? Different models of the same system → same behaviour under the same input?

Why attempts fail

- Incomplete description: missing parameters, initial conditions, inputs/stimuli or equations
- Errors in published equations or parameter values — one wrong value can break a result
- Code not shared, or tool/language not named
- Software and package versions not stated → main cause of failure even when code is available
- Unstandardised data analysis (e.g. burst detection in network models) → analysis code must be shared too
- Pathway figures showing components that are not actually in the model
- Exchange formats imperfect in practice (most tools failed to import the same SBML model)

Key findings

- Different simulation tools give the same result for the same model → the tools are not the bottleneck
- Models of the same plasticity in the same neuron type can respond differently to an identical input, while models of different brain areas can agree → models need a common comparison setup (same inputs, same readouts)
- Complexity predicts difficulty: simple models described in text were often fully reproduced; complex ones often failed even with parameter tables; replication from shared code mostly succeeded quickly

Good practice

- All equations, parameters, initial conditions and stimuli in tables

- Code in a public repository (e.g. ModelDB, BioModels) with tool and package versions
- Standard model formats (SBML, CellML, NeuroML, SED-ML) and minimum-information standards
- Reviewers rerun the code; journals publish reproduction studies (ReScience)
- Validate against experimental data and earlier models; state what the model adds